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Lecture – 11 : FACTORIZATION, EXPONENTIATION AND LOGARITHM

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FACTORIZATION

Fundamental Theorem of Arithmetic

$$n = p_1^{e1} \times p_2^{e2} \times \cdots \times p_k^{ek}$$

- Greatest Common Divisor

$$a = p_1^{a_1} \times p_2^{a_2} \times \cdots \times p_k^{a_k} \qquad b = p_1^{b_1} \times p_2^{b_2} \times \cdots \times p_k^{b_k}$$

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} \times p_2^{\min(a_2, b_2)} \times \cdots \times p_k^{\min(a_k, b_k)}$$

- Least Common Multiplier

$$a = p_1^{a1} \times p_2^{a2} \times \cdots \times p_k^{ak} \qquad b = p_1^{b1} \times p_2^{b2} \times \cdots \times p_k^{bk}$$

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \times p_2^{\max(a_2, b_2)} \times \cdots \times p_k^{\max(a_k, b_k)}$$

$$\text{lcm}(a, b) \times \gcd(a, b) = a \times b$$

Factorization methods

- Trial Division Method

```
Trial_Division_Factorization (n)          // n is the number to be factored
{
    a ← 2
    while (a ≤ √n )
    {
        while (n mod a = 0)
        {
            output a                      // Factors are output one by one
            n = n / a
        }
        a ← a + 1
    }
    if (n > 1) output n                   // n has no more factors
}
```

FACTORIZATION

- Example
  - Use the trial division algorithm to find the factors of 1233.
- Solution
  - We run a program based on the algorithm and get the following result.

$$1233 = 3^2 \times 137$$

FACTORIZATION

- Example
  - Use the trial division algorithm to find the factors of 1523357784
- Solution
  - We run a program based on the algorithm and get the following result.

$$1523357784 = 2^3 \times 3^2 \times 13 \times 37 \times 43987$$

Fermat Method

$$n = x^2 - y^2 = a \times b \qquad \text{with a} = (\text{x} + \text{y}) \text{ and b} = (\text{x} - \text{y})$$

```
Feramat_Factorization (n)          // n is the number to be factored
{
    x ← √n                         // smallest integer greater than √n
    while (x < n)
    {
    w ← x² − n
    if (w is perfect square)  y ← √w;  a ← x+y;  b ← x−y;   return a and b
    x ← x + 1
    }
}
```

FACTORIZATION

- More methods
  - Pollard p-1
  - Pollard rho
  - Number Field Sieve
  - Quadratic Sieve

Tutorial

CHINESE REMAINDER THEOREM

Used to solve a set of congruent equations with one variable but different moduli, which are relatively prime

$$x \equiv a_1 \pmod{m_1}$$
$$x \equiv a_2 \pmod{m_2}$$
$$\cdots$$
$$x \equiv a_k \pmod{m_k}$$

CHINESE REMAINDER THEOREM

- Example
  - The following is an example of a set of equations with different moduli:

$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 5$$
$$x \equiv 2 \pmod 7$$

  - The solution to this set of equations is given in the next section; for the moment, note that the answer to this set of equations is x = 23. This value satisfies all equations: 23 ≡ 2 (mod 3), 23 ≡ 3 (mod 5), and 23 ≡ 2 (mod 7).

CHINESE REMAINDER THEOREM

- Solution To Chinese Remainder Theorem
  - Find $M = m_1 \times m_2 \times \ldots \times m_k$. This is the common modulus.
  - Find $M_1 = M/m_1$, $M_2 = M/m_2$, …, $M_k = M/m_k$.
  - Find the multiplicative inverse of M_1, M_2, …, M_k using the corresponding moduli (m_1, m_2, …, m_k). Call the inverses M_1^{-1}, M_2^{-1}, …, M_k^{-1}.
  - The solution to the simultaneous equations is

$$x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1} + \cdots + a_k \times M_k \times M_k^{-1}) \bmod M$$

CHINESE REMAINDER THEOREM

- Example
  - Find the solution to the simultaneous equations:

$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 5$$
$$x \equiv 2 \pmod 7$$

- Solution: We follow the four steps.

1. $M = 3 \times 5 \times 7 = 105$

2. $M_1 = 105 / 3 = 35$, $M_2 = 105 / 5 = 21$, $M_3 = 105 / 7 = 15$

3. The inverses are $M_1^{-1} = 2$, $M_2^{-1} = 1$, $M_3^{-1} = 1$

4. $x = (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \bmod 105 = 23 \bmod 105$

CHINESE REMAINDER THEOREM

- Example
  - Find an integer that has a remainder of 3 when divided by 7 and 13, but is divisible by 12.
- Solution ????

CHINESE REMAINDER THEOREM

- Example
  - Find an integer that has a remainder of 3 when divided by 7 and 13, but is divisible by 12.
- Solution
  - This is a CRT problem. We can form three equations and solve them to find the value of x.

$$x = 3 \bmod 7$$
$$x = 3 \bmod 13$$
$$x = 0 \bmod 12$$

  - If we follow the four steps, we find x = 276. We can check that
    276 = 3 mod 7, 276 = 3 mod 13 and 276 is divisible by 12 (the quotient is 23 and the remainder is zero).

CHINESE REMAINDER THEOREM

- Assume we need to calculate z = x + y where x = 123 and y = 334, but our system accepts only numbers less than 100. These numbers can be represented as follows:

$$\begin{array}{ll} x \equiv 24 \pmod{99} & y \equiv 37 \pmod{99} \\ x \equiv 25 \pmod{98} & y \equiv 40 \pmod{98} \\ x \equiv 26 \pmod{97} & y \equiv 43 \pmod{97} \end{array}$$

- Adding each congruence in x with the corresponding congruence in y gives

$$\begin{array}{lll} x + y \equiv 61 \pmod{99} & \rightarrow & z \equiv 61 \pmod{99} \\ x + y \equiv 65 \pmod{98} & \rightarrow & z \equiv 65 \pmod{98} \\ x + y \equiv 69 \pmod{97} & \rightarrow & z \equiv 69 \pmod{97} \end{array}$$

- Now three equations can be solved using the Chinese remainder theorem to find z. One of the acceptable answers is z = 457.

CHINESE REMAINDER THEOREM

Secret Sharing scheme in cryptography aims to distribute and later recover secret S among n parties. Secret S is distributed in form of shares which are generated from secret. Without cooperation of k no. of parties, the secret cannot be reconstructed from shares directly. Consider the following example:

Say our secret is S. The shares for n=4 no. of parties are generated taking modulus 11,13,17 and 19. They are respectively 1,12,2 and 3 and given by following equations:

$S \equiv 1 \bmod 11,$

$S \equiv 12 \bmod 13,$

$S \equiv 2 \bmod 17,$

$S \equiv 3 \bmod 19.$

Now, from four possible sets of k=3 shares (as k shares are necessary to reconstruct the secret), consider one possible set {1, 12, 2} and recover the secret S from it.

CHINESE REMAINDER THEOREM

Solution: The problem can be solved by Chinese remainder theorem.

For the set {1,12,2}, the equations available are,

$S \equiv 1 \bmod 11$,

$S \equiv 12 \bmod 13$,

$S \equiv 2 \bmod 17$,

Now solving this equation using CRT, M=11 *13*17 = 2431,

M1 = 2431/11=221,

M2 = 2431/13=187,

M3=2431/17=143

$M1^{-1}$, $M2^{-1}$ and $M3^{-1}$ can be calculated using Extended Euclidean Algorithm.

$M1^{-1} = 1$

$M2^{-1} = 8$

$M3^{-1} = 5$

Now, secret S= ((1*221*1) + (12*187*8) + (2*143*5)) mod 2431

S = 155 mod 2431

EXPONENTIATION AND LOGARITHM

EXPONENTIATION AND LOGARITHM

Exponentiation: $y = a^x$ $\rightarrow$ **Logarithm:** $x = \log_a y$

Exponentiation

- Fast Exponentiation
  – The idea behind the square-and-multiply method

$$x = x_{n_b-1} \times 2^{k-1} + x_{n_b-2} \times 2^{k-2} + \cdots + x_2 \times 2^2 + x_1 \times 2^1 + x_0 \times 2^0$$

$$y = a^{x_{n_b-1} \times 2^{n_b-1} + x_{n_b-2} \times 2^{n_b-2} + \cdots + x_1 \times 2^1 + x_0 \times 2^0}$$

in which x_i is 0 or 1

$$y = \left[a^{2^{n_b-1}} \text{ or } 1\right] \times \left[a^{2^{n_b-2}} \text{ or } 1\right] \times \cdots \times \left[a^2 \text{ or } 1\right] \times \left[a \text{ or } 1\right]$$

Example:

$$y = a^9 = a^{1001_2} = a^8 \times 1 \times 1 \times a$$

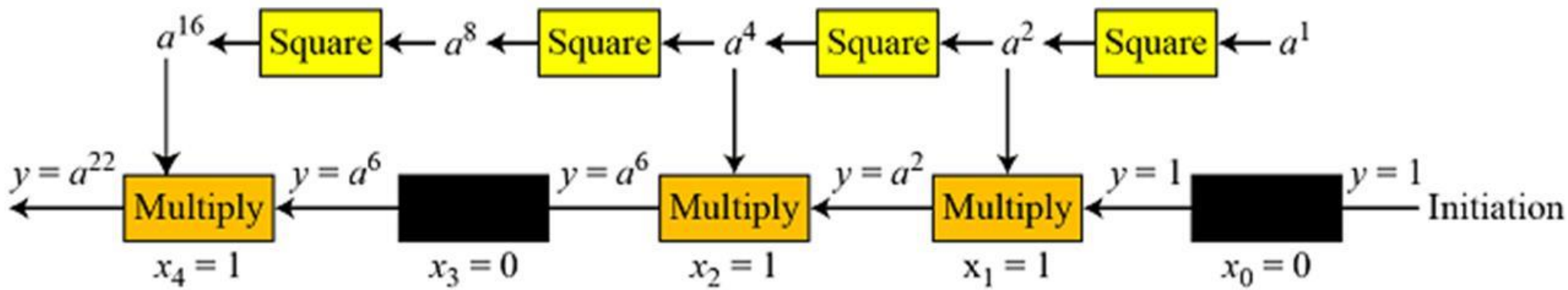
Exponentiation

```
Square_and_Multiply (a, x, n)
{
    y ← 1
    for (i ← 0 to n_b − 1)                  // n_b is the number of bits in x
    {
        if (x_i = 1)    y ← a × y mod n     // multiply only if the bit is 1

        a ← a^2 mod n                       // squaring is not needed in the last iteration
    }
    return y
}
```

Exponentiation

- The process for calculating $y = a^x$
- In this case, $x = 22 = (10110)_2$ in binary.



Exponentiation

Calculation of $17^{22} \bmod 21$

i	x_i	*Multiplication (Initialization: $y = 1$)*		*Squaring (Initialization: $a = 17$)*
0	0		$\rightarrow$	$a = 17^2 \bmod 21 = 16$
1	1	$y = 1 \times 16 \bmod 21 = 16$	$\rightarrow$	$a = 16^2 \bmod 21 = 4$
2	1	$y = 16 \times 4 \bmod 21 = 1$	$\rightarrow$	$a = 4^2 \bmod 21 = 16$
3	0		$\rightarrow$	$a = 16^2 \bmod 21 = 4$
4	1	$y = 1 \times 4 \bmod 21 = 4$	$\rightarrow$	

Logarithm

- In cryptography we need to discuss modular logarithm

Exhaustive search for modular logarithm

```
Modular_Logarithm (a, y, n)
{
    for (x = 1 to n −1)                          // k is the number of bits in x
    {
        if (y ≡ a^x mod n) return x
    }
    return failure
}
```

Logarithm

- Order of the Group.
- Example:
  - What is the order of group G = $<Z_{21}*, \times>$?
    - $|G| = \phi(21) = \phi(3) \times \phi(7) = 2 \times 6 = 12$. There are 12 elements in this group: 1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, and 20. All are relatively prime with 21.

Logarithm

- The order of an element, a, is the smallest integer i such that $a^i \equiv e \pmod{n}$.

  - Example:
    - Find the order of all elements in G = $<Z_{10}^*, \times>$.
    - This group has only $\phi(10) = 4$ elements: 1, 3, 7, 9.

The order of an element divides the order of the group (Lagrange theorem).

a. $1^1 \equiv 1 \bmod (10) \rightarrow \text{ord}(1) = 1.$
b. $3^1 \equiv 3 \bmod (10); 3^2 \equiv 9 \bmod (10); 3^4 \equiv 1 \bmod (10) \rightarrow \text{ord}(3) = 4.$
c. $7^1 \equiv 7 \bmod (10); 7^2 \equiv 9 \bmod (10); 7^4 \equiv 1 \bmod (10) \rightarrow \text{ord}(7) = 4.$
d. $9^1 \equiv 9 \bmod (10); 9^2 \equiv 1 \bmod (10) \rightarrow \text{ord}(9) = 2.$

Logarithm

- Primitive roots
  - In the group G = $<Z_n*, \times>$, when the order of an element is the same as ϕ(n), that element is called the primitive root of the group.
  - Example
    - There are no primitive roots in G = $<Z_8*, \times>$ because no element has the order equal to $\phi(8) = 4$.

Logarithm

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$
$a = 1$	**x: 1**	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1
$a = 3$	x: 3	**x: 1**	x: 3	x: 1	x: 3	x: 1	x: 3
$a = 5$	x: 5	**x: 1**	x: 5	x: 1	x: 5	x: 1	x: 5
$a = 7$	x: 7	**x: 1**	x: 7	x: 1	x: 7	x: 1	x: 7

The first time when x is 1, the value of i gives us the order of the element (double-sided boxes). The orders of elements are ord(1) = 1, ord(3) = 2, ord(5) = 2, and ord(7) = 2.

Logarithm

- Example
  - the result of $a^i \equiv x \pmod 7$ for the group $G = \langle Z_7^*, \times \rangle$. In this group, $\phi(7) = 6$.

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$
$a = 1$	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1
$a = 2$	x: 2	x: 4	x: 1	x: 2	x: 4	x: 1
Primitive root → $a = 3$	x: 3	x: 2	x: 6	x: 4	x: 5	x: 1
$a = 4$	x: 4	x: 2	x: 1	x: 4	x: 2	x: 1
Primitive root → $a = 5$	x: 5	x: 4	x: 6	x: 2	x: 3	x: 1
$a = 6$	x: 6	x: 1	x: 6	x: 1	x: 6	x: 1

The orders of elements are ord(1) = 1, ord(2) = 3, ord(3) = 6, ord(4) = 3, ord(5) = 6, and ord(6) = 2.

Therefore, this group has only two primitive roots: 3 and 5.

Logarithm

The group $G = \langle Z_n^*, \times \rangle$ has primitive roots only if n is 2, 4, p^t, or $2p^t$.

If the group $G = \langle Z_n^*, \times \rangle$ has any primitive root, the number of primitive roots is $\phi(\phi(n))$.

The group $G = \langle Z_n^*, \times \rangle$ is a cyclic group if it has primitive roots.
The group $G = \langle Z_p^*, \times \rangle$ is always cyclic.

The idea of Discrete Logarithm

Properties of $G = \langle Z_p^*, \times \rangle$:

1. Its elements include all integers from 1 to $p - 1$.
2. It always has primitive roots.
3. It is cyclic. The elements can be created using g^x where x is an integer from 1 to $\phi(n) = p - 1$.

4. The primitive roots can be thought as the base of logarithm. If the group has k primitive roots, calculations can be done in k different bases. Given $x = \log_g y$ for any element y in the set, there is another element x that is the log of y in base g. This type of logarithm is called **discrete logarithm**. A discrete logarithm is designated by several different symbols in the literature, but we will use the notation L_g to show that the base is g (the modulus is understood).

Solution to Modular Logarithm Using Discrete Logs

Tabulation of Discrete Logarithms

Discrete logarithm for $\mathbf{G} = \langle \mathbf{Z}_7^*, \times \rangle$

y	1	2	3	4	5	6
$x = L_3\, y$	6	2	1	4	5	3
$x = L_5\, y$	6	4	5	2	1	3

- Find x in each of the following cases:

  a. $4 \equiv 3^x \pmod 7$

  b. $6 \equiv 5^x \pmod 7$

- Solution
  - Use the tabulation of the discrete logarithm

    a. $4 \equiv 3^x \bmod 7 \rightarrow x = L_3 4 \bmod 7 = 4 \bmod 7$

    b. $6 \equiv 5^x \bmod 7 \rightarrow x = L_5 6 \bmod 7 = 3 \bmod 7$

Logarithm

Using Properties of Discrete Logarithms

Traditional Logarithm	*Discrete Logarithms*
$\log_a 1 = 0$	$L_g 1 \equiv 0 \pmod{\phi(n)}$
$\log_a (x \times y) = \log_a x + \log_a y$	$L_g(x \times y) \equiv (L_g x + L_g y) \pmod{\phi(n)}$
$\log_a x^k = k \times \log_a x$	$L_g x^k \equiv k \times L_g x \pmod{\phi(n)}$

The discrete logarithm problem has the same complexity as the factorization problem.